Lakshya JEE 2.0 2025

Mathematics

DPP: 2

Indefinite Integration

Q1 $\int \frac{1-\tan^2 x}{1+\tan^2 x} dx$

Q2 $\int \frac{\sin 2x+\sin 5x-\sin 3x}{\cos x+1-2\sin^2 2x} dx =$

(A) $2\cos x + c$

(B) $-2\sin x + c$

(C) $2\sin x + c$

(D) $-2\cos x + c$

Q3 $\int \frac{x^4+3x^2+1}{\sqrt{x}} dx$ equals

(A) $\frac{2}{7}x^{\frac{7}{2}} + 2x^{\frac{3}{2}} + x^{-\frac{1}{2}} + C$

(B) $\frac{2}{9}x^{\frac{7}{2}} + \frac{2}{3}x^{\frac{3}{2}} + \sqrt{x} + C$

(C) $\frac{2}{5}x^{\frac{5}{2}} + \frac{3}{2}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C$

(D) $\frac{2}{9}x^{\frac{9}{2}} + \frac{6}{5}x^{\frac{5}{2}} + 2x^{\frac{1}{2}} + C$

Q4 $\int \frac{2+3x^2}{x^2(1+x^2)} dx =$

(A) $\tan^{-1} x + \frac{2}{x} + c$

(B) $\frac{2}{x} - \tan^{-1} x + c$

(C) $\tan^{-1} x - \frac{1}{x} + c$

(D) $\tan^{-1} x - \frac{2}{x} + c$

Q5 $\int \frac{x^4+x^2+1}{2(1+x^2)} dx$

Q6 $\int \frac{x^3}{x+1} dx$

Q7 $\int \frac{\sin^6 x+\cos^6 x}{\sin^2 x\cos^2 x} dx$

Q8 $\int \frac{(x^2+\sin^2 x)\sec^2 x}{1+x^2} dx =$

(A) $\tan x + \tan^{-1} x + c$

(B) $\tan^{-1} x - \tan x + c$

(C) $\tan x - \tan^{-1} x + c$

(D) $c - \tan x - \tan^{-1} x$

Q9 $\int \sin 3x \cos x dx$

Q10 $\int \frac{dx}{\sin^2 x\cos^2 x}$

Q11 $\int \sin^4 x dx$

Q12 $\int \frac{dx}{1+\sin x} = c - f(x)$, then $f(x)$ equals:

(A) $\tan x - \sec x$

(B) $\sec x - \tan x$

(C) $\sec x + \tan x$

(D) $-\sec x - \tan x$

Q13 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$\int 4\cos\frac{x}{2}.\cos x.\sin\frac{21}{2}x dx$

Q14 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{\cos x-\sin x}{\cos x+\sin x}(2+2\sin 2x) dx$$

Q15 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$\int \left(3\sin x\cos^2 x - \sin^3 x\right) dx$

Q16 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$\int \frac{e^{2x}-1}{e^x} dx$

Q17 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$\int \frac{\sin 2x+\sin 5x-\sin 3x}{\cos x+1-2 \sin^2 2x} dx$

Q18 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \left[\frac{\cot^2 2x-1}{2 \cot 2x} - \cos 8x \cot 4x\right] dx$

Q19 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{\cos^4 x-\sin^4 x}{\sqrt{1+\cos 4x}} dx (\cos 2x>0)$$

Q20 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{\cos 8x-\cos 7x}{1+2 \cos 5x} dx$

Q21 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{\cos 4x+1}{\cot x-\tan x} dx$

Q22 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \left[\sin^2\left(\frac{9\pi}{8}+\frac{x}{4}\right)-\sin^2\left(\frac{7\pi}{8}+\frac{x}{4}\right)\right] dx$

Q23 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \tan^{-1}\left(\frac{1+\cos x}{\sin x}\right) dx$$

Q24 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{x^6-1}{x^2+1} dx$

Q25 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int sinx\ cosx\ cos(2x)\ cos(4x)\ dx$

Q26 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.

$$\int \frac{2x-1}{x-2} dx$$

Q27 Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying.
$\int \frac{2x^3+3x^2+4x+5}{2x+1} dx$

Answer Key

Q1 $\frac{1}{2}\sin 2x + C$

Q2 (D)

Q3 (D)

Q4 (D)

Q5 $\frac{1}{2}\left[\frac{x^3}{3} + \tan^{-1} x\right] + C$

Q6 $\frac{x^3}{3} - \frac{x^2}{2} + x - \log|x+1| + c$

Q7 $\tan x - cot x - 3x + C$

Q8 (C)

Q9 $\frac{-\cos 4x}{8} - \frac{\cos 2x}{4} + C$

Q10 $-2\cot 2x + c$

Q11 $\frac{3x}{8} - \frac{\sin 2x}{4} + \frac{\sin 4x}{32} + C$

Q12 (B)

Q13 $-\left[\frac{1}{9}\cos 9x + \frac{1}{10}\cos 10x + \frac{1}{11}\cos 11x + \frac{1}{12}\cos 12x\right] + C$

Q14 $sin 2x + c$

Q15 $-\frac{\cos 3x}{3} + C$

Q16 $e^x + e^{-x} + C$

Q17 - 2 cos x + C

Q18 -(cos 8x)/8 +C

Q19 $\frac{x}{\sqrt{2}} + C$

Q20 $\frac{\sin 3x}{3} - \frac{\sin 2x}{2} + C$

Q21 $-\frac{\cos 4x}{8} + C$

Q22 $-\sqrt{2}\cos\frac{x}{2} + C$

Q23 $\frac{\pi x}{2} - \frac{x^2}{4} + C$

Q24 $\frac{x^5}{5} + x - \frac{x^3}{3} - 2\tan^{-1} x + c$

Q25 $-\frac{1}{64}\cos 8x + C$

Q26 $2x + 3\ln(x-2) + C$

Q27 $\frac{x^3}{3} + \frac{x^2}{2} + \frac{3x}{2} + \frac{7}{4}\ln(2x+1)$ + C

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